

WEEK 2 – DATA MODELING, DAX & SALES DASHBOARD

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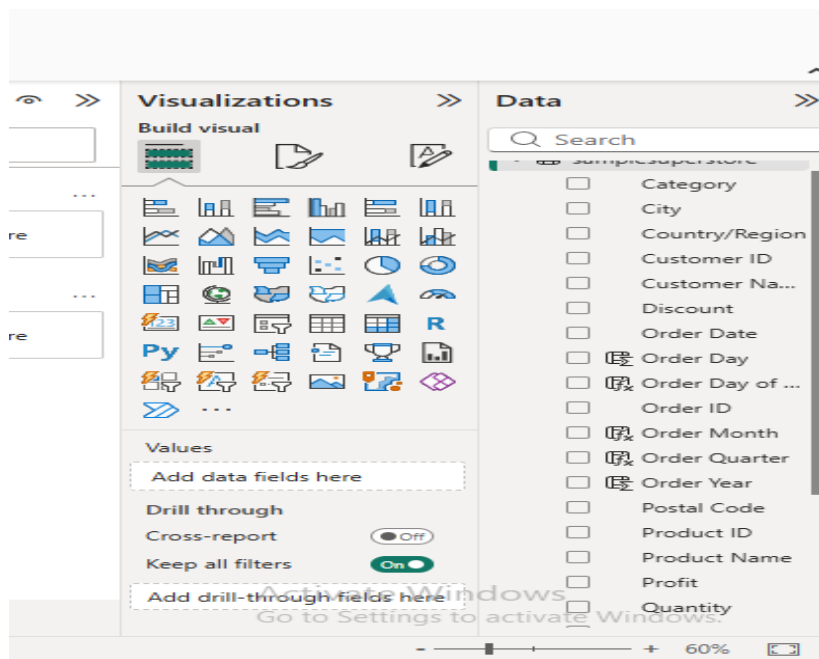
COURSE: BSc DS&AI

INTRODUCTION:

The aim of this project is to create an interactive Superstore Sales Dashboard using Microsoft Power BI. The project focuses on data modeling, DAX calculations, relationships between tables, and interactive data visualization. The dashboard helps to analyze sales, profit, orders, quantity, categories and regional performance.

DATASET:

The Superstore Sales Dataset contains information about customers, orders, products, sales, profit, quantity, categories and regions. The dataset was imported into Power BI and used for creating the sales dashboard



CREATING THE MEASURES TABLE

A separate table was created to store the DAX measures. This helps organize all the calculations in one place and makes the Power BI model easier to manage.

Create Table □ ×

	Column1	+
1	Measures	
2	Measures	
+		

Name:

Load

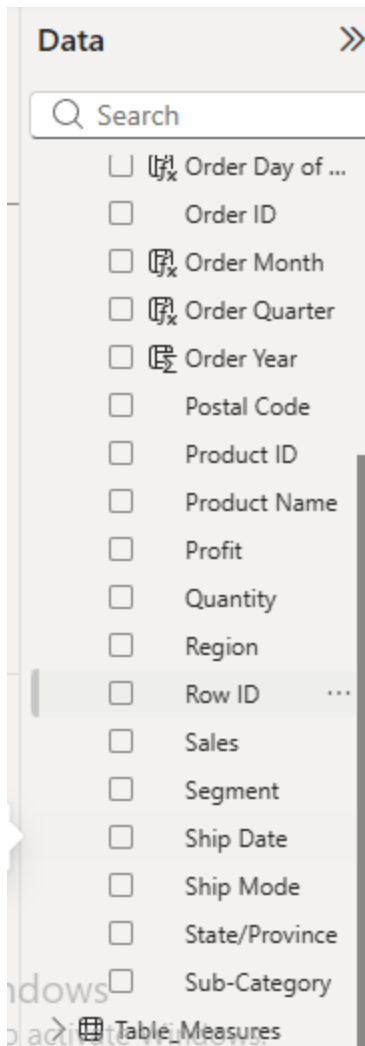
Edit

Cancel

Activ
Go to

RENAMING THE MEASURES TABLE

The measures table was organized and named `_Measures`. This table was used to store the calculations required for the dashboard.



CREATING DAX MEASURES

DAX measures were created to calculate the important business metrics required for the dashboard.

Total Sales

Total Sales =
SUM(samplesuperstore[Sales])

Total Profit

Total Profit =
SUM(samplesuperstore[Profit])

Total Orders

Total Orders =
`DISTINCTCOUNT(samplesuperstore[Order ID])`

Total Quantity

Total Quantity =
`SUM(samplesuperstore[Quantity])`

Average Sales

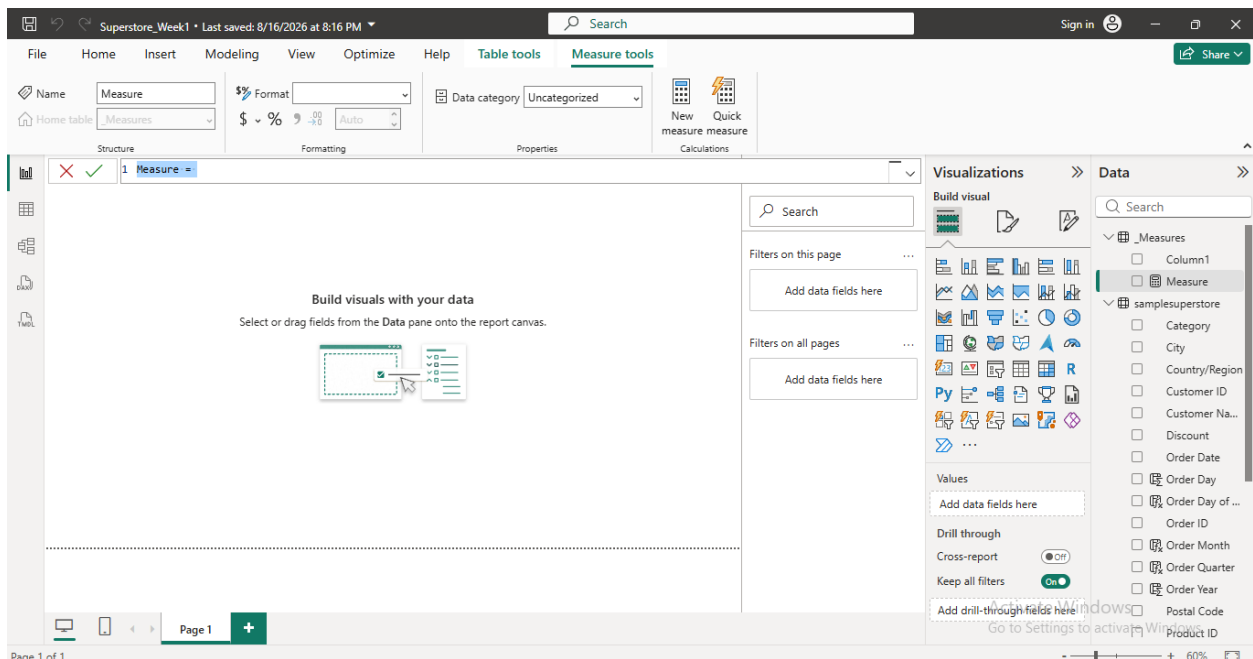
Average Sales =
`AVERAGE(samplesuperstore[Sales])`

Profit Margin %

Profit Margin % =
`DIVIDE([Total Profit], [Total Sales], 0)`

CREATING A MEASURE

Measures were created using the New Measure option in Power BI. DAX expressions were entered in the formula bar to calculate the required metrics



CREATING THE DATE TABLE

A separate date table named DimDate was created to support time-based analysis. The table contains a continuous range of dates based on the minimum and maximum Order Date values.

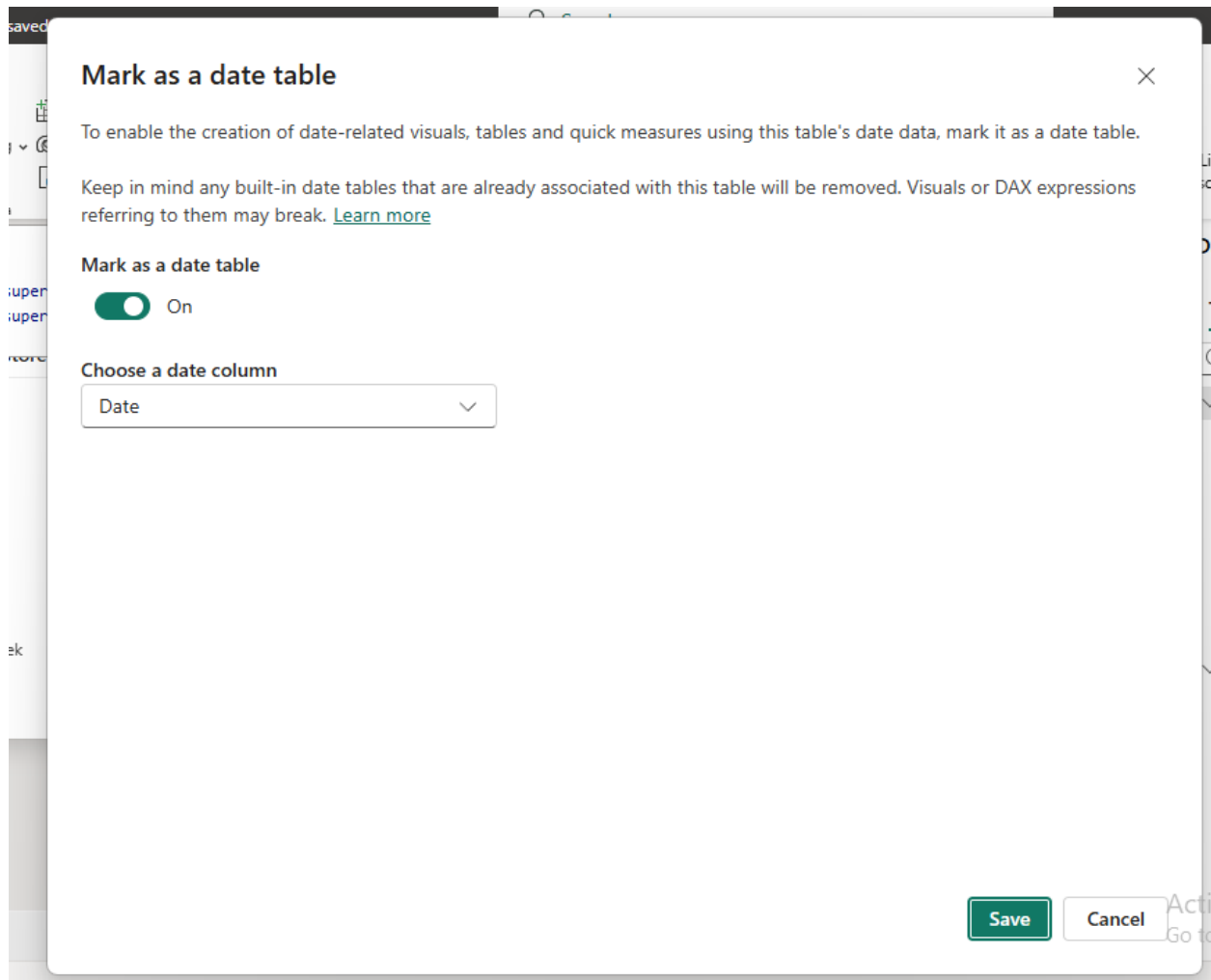
DAX:

DimDate =

```
CALENDAR(  
    MIN(samplesuperstore[Order Date]),  
    MAX(samplesuperstore[Order Date])  
)
```

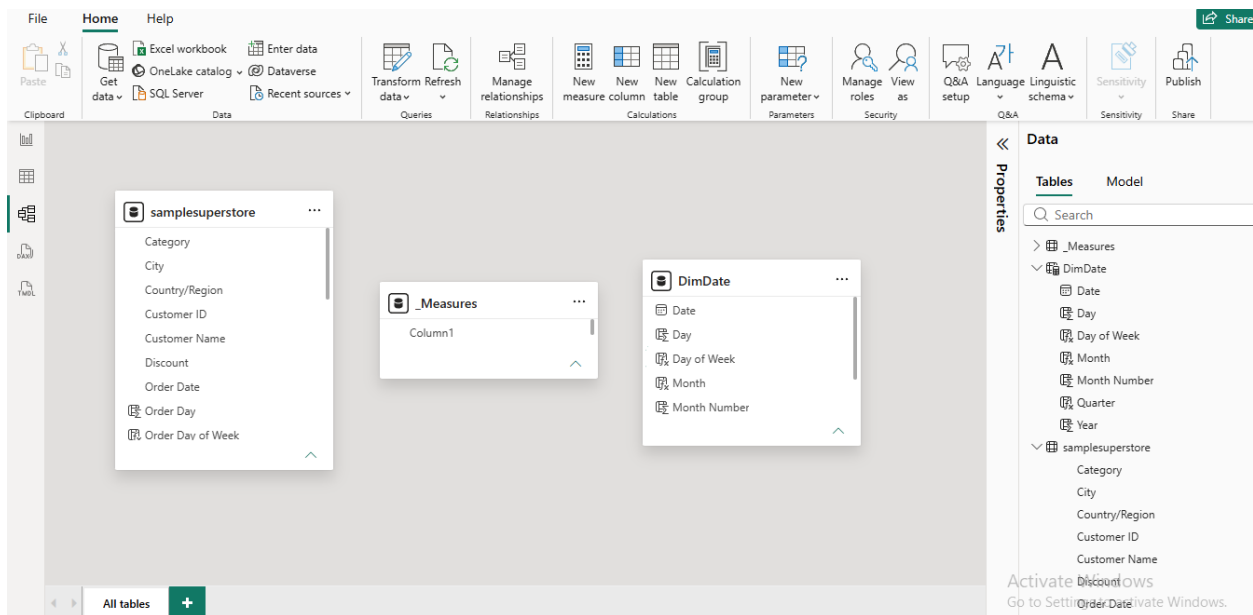
MARKING DIMDATE AS DATE TABLE

The DimDate table was marked as a date table using the Date column. This allows Power BI to recognize it as a proper date dimension



DATA MODEL

The Power BI model contains the main Superstore sales table, the DimDate table and the dedicated _Measures table. The date table is connected to the sales table to support time-based analysis.



KPI CARDS

KPI cards were used to display the important business performance indicators in a simple and easily understandable format. The measures include Total Sales, Total Profit, Total Orders, Total Quantity, Average Sales and Profit Margin.

Establishing Table Relationships

Relationships were created between the tables to build the Power BI data model.

The main relationship used is:

DimDate[Date] → samplesuperstore[Order Date]

The relationship uses:

- **Cardinality:** One to many (1:*)
- **Cross-filter direction:** Single
- **Active relationship:** Yes

The screenshot displays the Microsoft Power BI Desktop interface. The top ribbon shows various tabs: Clipboard, Data, Queries, Relationships, Calculations, Parameters, and Security. The 'Data' tab is active, showing a list of tables in the 'samplesuperstore' database. The tables listed are: City, Country/Region, Customer ID, Customer Name, Discount, Order Date, Order Day, Order Day of Week, and Order ID. The '_Measures' table is highlighted, and a tooltip is visible showing its properties: Name '_Measures', Storage mode 'Import', and Data refreshed '9/8/2026, 9:14:57 AM'. The 'Average Sales' measure is shown with the formula 'SUM(Sales[Sales Amount])'. The 'Order Date' table is also visible, with a tooltip showing its properties: Name 'Order Date', Storage mode 'Import', and Data refreshed '9/8/2026, 9:14:57 AM'. The 'Order Date' table is connected to the '_Measures' table via a relationship line.

The relationship between the date table and sales table was configured as a **one-to-many relationship**.

This structure follows the basic concept of a **star schema**, where the main sales table is connected to supporting dimension tables.

New relationship

Select tables and columns that are related.

From table

DimProduct

Category	Product ID	Product Name	Sub-Category
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A preview of this table isn't available

To table

samplesuperstore

Category	City	Country/Regi...	Customer ID	Customer Na...	Discount	Order Dat
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Cardinality

One to many (1:*)

Cross-filter direction

Single

☒ Make this relationship active

☐ Apply security filter in both directions

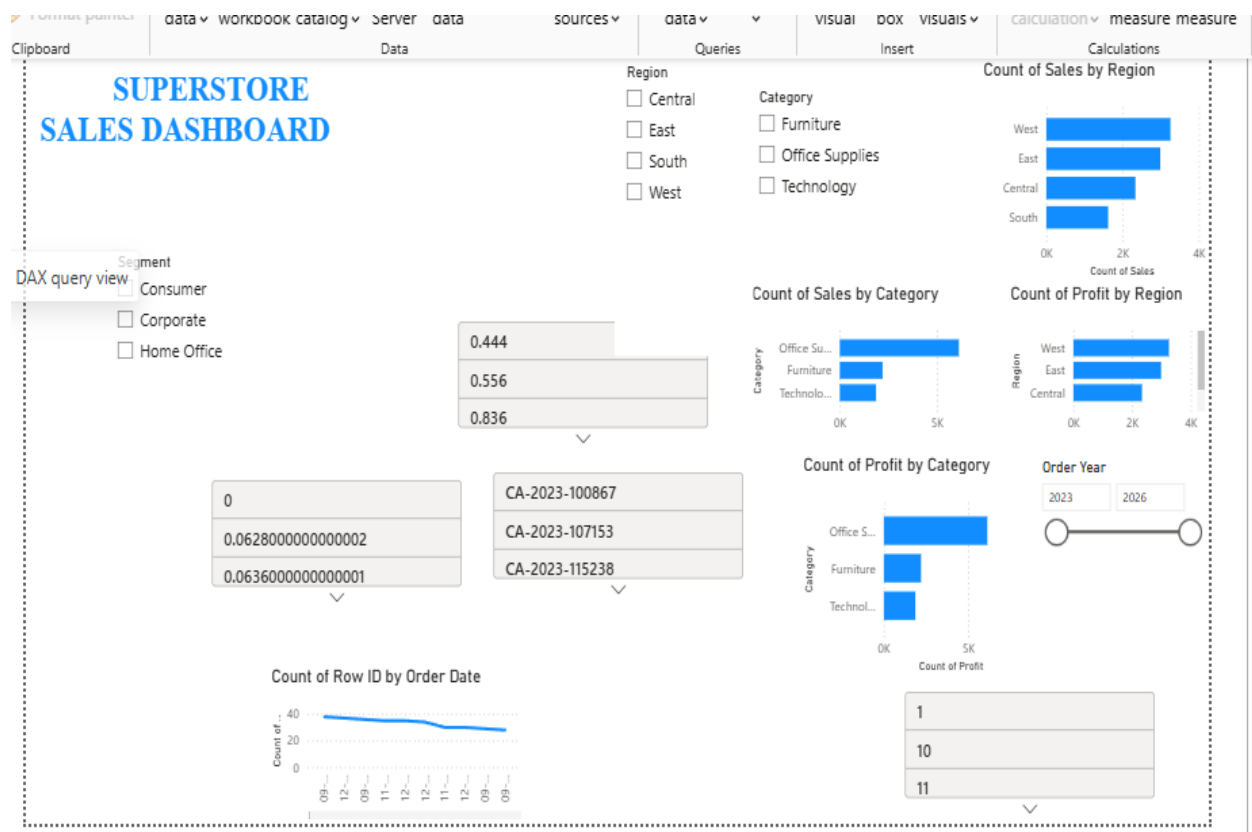
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Model Validation

After establishing the relationship, the model was checked in Model View to ensure that the tables were connected correctly.

The relationship between the date table and the Superstore sales table enables filtering and analysis based on dates.



Interactive Dashboard Preparation

The created DAX measures and data model provide the foundation for creating an interactive sales dashboard. The measures can be used for KPI cards and other visuals, while the Dim Date relationship supports time-based analysis.

Conclusion

The Week 2 Power BI project demonstrates the process of preparing a structured data model, creating DAX measures and establishing relationships between tables. A dedicated measures table was created to organize calculations, while the DimDate table was created

for date-based analysis. The relationships between the tables provide a proper foundation for developing an interactive Superstore Sales Dashboard.